Model sketch: multi-sector New Keynesian economy with an endogenous production network

This document builds the model from primitives. Every assumption is listed once (Section 1) with a label, every actor's problem is written and solved (Section 2), the flex-price equilibrium is derived exactly (Section 3), the network-choice block is derived from the firm's problem including a re-derivation of KMNT's Lemma 2 for monopolistic producers (Section 4), the sticky-price block is log-linearized step by step and mapped onto R23's objects (Section 5), inconsistencies and undesired effects are listed and mapped to the assumptions that cause them (Section 6), and a notation dictionary closes (Section 7). All derivations are in the main text.

Conventions: lower-case letters are logs of the corresponding upper-case variables ($p_i = \log P_i$); vectors are column vectors indexed over sectors $i = 0, 1, \ldots, N$; $\mathbf{1}$ is the vector of ones; e_0 is the first unit vector; $\mathbb{E}_t$ is the expectation conditional on period-t information; a superscript *nat* denotes the flexible-price ("natural") value; a tilde denotes a gap from natural. Sector 0 is the labor union.

0. Overview and timing

There are three nested clocks.

1. **Regime.** A regime is a belief pair (μ, Σ) describing the stationary distribution of sectoral productivity. A fragmentation shock is a once-and-for-all change of regime.
2. **Network.** At the start of a regime, each sector chooses its input technique Ω_i once, before any productivity is realized, on the basis of (μ, Σ). The technique stays fixed for the life of the regime.
3. **Period.** Every period t, productivity ε_t is realized, a fraction of producers reset prices (Calvo), the central bank sets the nominal rate, and markets clear.

The two blocks of the model correspond to clocks 2 and 3. Because the technique is evaluated under the flexible-price distribution (assumption I2 below), the network block is solved first and its output — the input-output matrix Ω^*, the Domar weights λ^*, the steady-state TFP levels $a_i(\Omega_i^*)$ — becomes the linearization point of the sticky-price block. The structure is recursive: no fixed point across blocks.

1. Assumptions

Each assumption carries a label used in Section 6 to trace consequences back to their source.

Environment

Label	Assumption
E1	Discrete time $t = 0, 1, 2, \ldots$, infinite horizon. Closed economy. No capital, no inventories, no government consumption.
E2	Sectors $i \in \{0, 1, \ldots, N\}$. Sector 0 is the labor union; sectors $1, \ldots, N$ produce goods.
E3	Sectoral productivity $\varepsilon_t = (\varepsilon_{1t}, \ldots, \varepsilon_{Nt})$ is a stationary Gaussian process with unconditional mean μ and unconditional covariance Σ; $\varepsilon_{0t} \equiv 0$ (the union has no shock). Realized log TFP of sector i is $\varepsilon_{it} + a_i(\Omega_i)$. Within a regime the deviation $\log A_{it} \equiv \varepsilon_{it} - \mu_i$ follows $\log A_t = \eta \log A_{t-1} + \xi_t$, $\xi_t \sim N(0, (1-\eta^2)\Sigma)$, so that $\mathbb{V}[\log A_t] = \Sigma$ unconditionally.
E4	A fragmentation shock is a regime change $(\mu, \Sigma) \to (\mu', \Sigma')$ concentrated on foreign-labelled sectors. Shocks to access (tariffs, export controls) are wasteful wedges $\log(1 + \tau^w)$ on the buyer's cost; by KMNT App. L Prop. 13(2) they enter exactly as ε does, so they are folded into ε.

Preferences

Label	Assumption
P1	Representative household with period utility $\dfrac{C_t^{1-\gamma}}{1-\gamma} - \dfrac{L_t^{1+\phi}}{1+\phi}$, discount factor $\rho \in (0,1)$, $\gamma > 0$, $\phi > 0$.
P2	Consumption aggregator is Cobb–Douglas, $C_t = \prod_{i=1}^{N} (C_{it}/\beta_i)^{\beta_i}$, $\sum_i \beta_i = 1$, $\beta_0 = 0$.
P3	$\gamma > 1$.
P4	Disutility depends on total hours L_t only, not on their distribution across labor types (pooling).

Technology

Label	Assumption
T1	Producer f in sector $i \geq 1$ has $Y_{ift} = e^{\varepsilon_{it} + a_i(\Omega_i)} \zeta_i \prod_{j=0}^{N} X_{ifjt}^{\Omega_{ij}}$ with $\sum_{j=0}^{N} \Omega_{ij} = 1$ (constant returns), $\zeta_i \equiv \prod_j \Omega_{ij}^{-\Omega_{ij}}$, where X_{if0t} is labor services bought from the union.
T2	The technique-dependent TFP shifter is quadratic, $a_i(\Omega_i) = \frac{1}{2}(\Omega_i - \Omega_i^\circ)^\top H_i (\Omega_i - \Omega_i^\circ) \leq 0$, with H_i symmetric negative definite and ideal shares Ω_i° in the interior of the simplex.
T3	One technique per sector, chosen once per regime and interior. All producers in a sector use the same technique in equilibrium.
T4	The union (sector 0) transforms raw household hours into labor services one-for-one: $Y_{0ft} = L_{ft}$. It uses no intermediates ($\Omega_{0j} = 0$).

Market structure

Label	Assumption
M1	Within each sector $i \in \{0, \ldots, N\}$ a continuum $f \in [0,1]$ of monopolistically competitive producers sells differentiated varieties aggregated with CES elasticity $\theta > 1$, common across sectors.
M2	Producers receive an input subsidy τ on all purchases with $(1-\tau)\,\mathcal{M} = 1$, $\mathcal{M} \equiv \theta/(\theta-1)$, financed by a lump-sum tax on the household. Profits are rebated to the household.
M3	Markets are complete: the household's marginal utility of nominal income is the state price used by all producers.
M4	"Foreign" is a label. A foreign sector differs from a domestic one only in (μ_f, Σ_{ff}) and in its covariances. Price rigidity is equal across foreign and domestic suppliers, $\delta_F = \delta_D$. Foreign sectors buy labor services from the domestic union like any other sector.

Nominal rigidity

Label	Assumption
N1	Calvo pricing: in each period a producer in sector i (including union members, $i = 0$) may reset its price with probability $\delta_i \in (0,1)$, independently across producers and time. Sector 0's reset probability $\delta_0 \equiv \delta_L < 1$ is wage rigidity.
N2	The economy is log-linearized around the zero-inflation flexible-price steady state of the regime, at which every price equals marginal cost and the network is Ω^*.

Information and timing

Label	Assumption
I1	Techniques are chosen at the start of a regime, before any ε_t of the regime is observed, using the unconditional distribution $N(\mu, \Sigma)$.
I2	Frequency separation: when evaluating a technique, producers use the distribution of prices, quantities and the discount factor that would obtain under flexible prices.
I3	All agents, including the central bank, observe Ω^* and hence λ^*.

Policy

Label	Assumption
G1	The central bank sets the nominal interest rate so that the output gap is zero every period, $\tilde{y}_t = 0$. (R23 Lemma 4 shows this is implementable by a Taylor rule on the divine-coincidence index.) To be relaxed to a Taylor rule on CPI in a later exercise.

2. Actors and their problems

2.1 Household

The household chooses consumption $\{C_{it}\}$, raw hours L_t and one-period nominal bonds B_t to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \rho^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\phi}}{1+\phi} \right]$$

subject to

$$\sum_{i=1}^{N} P_{it} C_{it} + B_t \le W_t^h L_t + (1 + i_{t-1}) B_{t-1} + \Pi_t - T_t, \qquad C_t = \prod_{i=1}^{N} \left(\frac{C_{it}}{\beta_i} \right)^{\beta_i},$$

where W_t^h is the price at which the household sells raw hours to the union, Π_t is aggregate profit (all sectors, including the union) and T_t the lump-sum tax.

Expenditure minimization. Minimizing $\sum_i P_{it} C_{it}$ subject to $\prod_i (C_{it}/\beta_i)^{\beta_i} = 1$ with multiplier Λ^C gives $P_{it} = \Lambda^C \beta_i / C_{it}$, hence $C_{it} = \Lambda^C \beta_i / P_{it}$. Substituting into the constraint, $\prod_i (\Lambda^C / P_{it})^{\beta_i} = 1$, so $\Lambda^C = \prod_i P_{it}^{\beta_i} \equiv P_t^C$. Therefore

$$C_{it} = \beta_i \frac{P_t^C C_t}{P_{it}}, \qquad P_t^C = \prod_{i=1}^{N} P_{it}^{\beta_i}, \qquad p_t^C = \beta^\top p_t. \tag{2.1}$$

The CPI is an exact geometric average; no log-linearization is needed.

Labor supply. The first-order condition for L_t with the multiplier on the budget constraint $\Lambda_t \rho^t$ gives $L_t^\phi = \Lambda_t W_t^h$, and the condition for C_t gives $C_t^{-\gamma} = \Lambda_t P_t^C$. Combining,

$$\frac{W_t^h}{P_t^C} = C_t^\gamma L_t^\phi \qquad \Longleftrightarrow \qquad w_t^h = p_t^C + \gamma c_t + \phi l_t. \tag{2.2}$$

Under P4 there is a single marginal rate of substitution for all hours, whatever labor type they are supplied as.

Euler equation. The condition for B_t, $\rho^t \Lambda_t = \rho^{t+1}(1 + i_t)\mathbb{E}_t \Lambda_{t+1}$, with $\Lambda_t = C_t^{-\gamma}/P_t^C$, is

$$1 = \rho(1 + i_t)\, \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} \frac{P_t^C}{P_{t+1}^C} \right]. \tag{2.3}$$

Stochastic discount factor. The nominal state price for a period-$t+s$ payoff, in units of period-t money, is

$$\Lambda_{t,t+s} = \rho^s \left(\frac{C_{t+s}}{C_t} \right)^{-\gamma} \frac{P_t^C}{P_{t+s}^C}. \tag{2.4}$$

By M3 this is what every producer uses.

2.2 Producers in sectors $i \geq 1$

Producer f in sector i solves three problems: an ex-post input choice every period, a Calvo price-reset problem whenever selected, and an ex-ante technique choice once per regime.

(a) Ex-post input choice. Given technique Ω_i, realized TFP and input prices, minimize $\sum_{j=0}^{N} P_{jt} X_{ifjt}$ subject to producing one unit. With multiplier Λ^X, the first-order condition for input j is $P_{jt} = \Lambda^X \Omega_{ij} / X_{ifjt}$ (using the constraint $Y = 1$), i.e. $X_{ifjt} = \Lambda^X \Omega_{ij} / P_{jt}$. Substituting into the production constraint,

$$e^{\varepsilon_{it}+a_i}\zeta_i \prod_j \left(\frac{\Lambda^X \Omega_{ij}}{P_{jt}}\right)^{\Omega_{ij}} = 1 \implies \Lambda^X = \frac{\prod_j P_{jt}^{\Omega_{ij}}}{e^{\varepsilon_{it}+a_i}\zeta_i \prod_j \Omega_{ij}^{\Omega_{ij}}} = \frac{\prod_j P_{jt}^{\Omega_{ij}}}{e^{\varepsilon_{it}+a_i(\Omega_i)}},$$

where the last step uses $\zeta_i \prod_j \Omega_{ij}^{\Omega_{ij}} = 1$. The multiplier is the pre-subsidy unit cost, common to all producers in the sector:

$$MC_{it} = \frac{\prod_{j=0}^{N} P_{jt}^{\Omega_{ij}}}{e^{\varepsilon_{it}+a_i(\Omega_i)}}, \qquad mc_{it} = \sum_{j=0}^{N} \Omega_{ij} p_{jt} - \varepsilon_{it} - a_i(\Omega_i), \tag{2.5}$$

and conditional factor demands are $X_{ifjt} = \Omega_{ij} MC_{it} Y_{ift} / P_{jt}$. Cost shares are constant and equal to Ω_{ij} (T1, T4): this is the Cobb–Douglas property that makes (2.5) exactly log-linear in prices and productivity. With the subsidy (M2) the producer's effective marginal cost is $(1-\tau)MC_{it} = MC_{it}/\mathcal{M}$.

(b) Price setting. Demand for variety f is $Y_{ift} = (P_{ift}/P_{it})^{-\theta} Y_{it}$ with sectoral index $P_{it} = \left(\int_0^1 P_{ift}^{1-\theta} df\right)^{1/(1-\theta)}$. A producer selected to reset in t (probability δ_i) chooses P_{it}^* to maximize the discounted value of profits over the horizon during which the price survives:

$$\max_{P^*} \; \mathbb{E}_t \sum_{s=0}^{\infty} (1-\delta_i)^s \Lambda_{t,t+s} \left[P^* - \frac{MC_{i,t+s}}{\mathcal{M}}\right] \left(\frac{P^*}{P_{i,t+s}}\right)^{-\theta} Y_{i,t+s}.$$

Differentiating with respect to P^* and dividing by $(1-\theta)(P^*)^{-\theta}$ gives:

$$P_{it}^* \;=\; \frac{\mathbb{E}_t \sum_{s=0}^{\infty} (1-\delta_i)^s \, \Lambda_{t,t+s} \, P_{i,t+s}^{\theta} \, Y_{i,t+s} \, MC_{i,t+s}}{\mathbb{E}_t \sum_{s=0}^{\infty} (1-\delta_i)^s \, \Lambda_{t,t+s} \, P_{i,t+s}^{\theta} \, Y_{i,t+s}} \tag{2.6}$$

The reset price is a weighted average of current and expected future *pre-subsidy* marginal costs; the markup and the subsidy cancel exactly (M2). This is what makes the flexible-price limit ($\delta_i \to 1$) efficient: $P_{it} = MC_{it}$.

By the law of large numbers across producers, the sectoral index evolves as

$$P_{it}^{1-\theta} = (1-\delta_i) P_{i,t-1}^{1-\theta} + \delta_i (P_{it}^*)^{1-\theta}. \tag{2.7}$$

(c) Technique choice. At the start of the regime, producer f chooses $\Omega_{if} \in \{\Omega \in \mathbb{R}_+^{N+1} : \mathbf{1}^\top \Omega = 1\}$ to maximize the expected discounted value of its profit stream, taking as given the sectoral price P_{it}, sectoral demand Y_{it} and the discount factor. Under I1–I2 this value is evaluated at the flexible-price distribution, in which the producer resets its price every period. Solved in Section 4.

2.3 The union (sector 0)

Union member f buys raw hours from the household at W_t^h and sells differentiated labor services at price $W_{ft} \equiv P_{0ft}$. Firms' labor demand aggregates the varieties with the same CES elasticity θ, so $W_t \equiv P_{0t} = \left(\int W_{ft}^{1-\theta} df\right)^{1/(1-\theta)}$ is the wage index paid by producers. Under T4 the union's marginal cost is $MC_{0t} = W_t^h$, i.e. $mc_{0t} = w_t^h$, and by (2.2)

$$mc_{0t} = p_t^C + \gamma c_t + \phi l_t. \tag{2.8}$$

The union member's reset problem is (2.6) with $i = 0$ and reset probability $\delta_0 = \delta_L$. Sector 0 is therefore formally identical to any other sector: it has a Calvo Phillips curve, a subsidy that offsets its markup, and a marginal cost given by (2.8) instead of (2.5). In the notation of Section 5, sector 0 is the only sector with a nonzero direct labor share: $\alpha \equiv e_0$, and all other sectors' labor shares appear in column 0 of Ω.

2.4 Government

The government pays the subsidy $\tau \sum_i \sum_j P_{jt} X_{ijt}$ and levies T_t on the household so that its budget balances period by period: $T_t = \tau \sum_i \sum_j P_{jt} X_{ijt}$. It issues no debt.

2.5 Central bank

Under G1 the central bank chooses i_t such that $\tilde{y}_t \equiv y_t - y_t^{nat} = 0$ for all t. Nothing else about the rule is needed in Section 5; R23 Lemma 4 supplies existence, uniqueness and an implementing Taylor rule. The Euler equation (2.3) is then needed only to back out i_t.

2.6 Market clearing

Goods: $Y_{it} = C_{it} + \sum_{j=1}^{N} \int_0^1 X_{jfit}\, df$ for $i \geq 1$. Labor services: $Y_{0t} = \sum_{j=1}^{N} \int_0^1 X_{jf0t}\, df$. Raw hours: $L_t = \int_0^1 Y_{0ft}\, df$. Bonds: $B_t = 0$.

2.7 Equilibrium

Given a regime (μ, Σ), an equilibrium is a network Ω^* and stochastic processes for prices, quantities and the interest rate such that (i) each producer's technique solves 2.2(c) given the flex-price distribution induced by Ω^* and the other producers' techniques; (ii) each period, given Ω^* and realized ε_t, input choices solve 2.2(a), reset prices solve (2.6), sectoral indices follow (2.7), the household solves 2.1, the government budget balances, the central bank implements G1, and markets clear. Under I2 the equilibrium is computed recursively: Section 4 solves (i) using the exact flex-price equilibrium of Section 3, and Section 5 solves (ii) to first order around it.

3. The flexible-price equilibrium at a given network (exact)

Fix a network Ω and suppose every producer can reset its price every period ($\delta_i = 1$ for all i). Then (2.6) gives $P_{it} = MC_{it}$ for all i, including $W_t = W_t^h$. Everything below holds exactly, not to first order, because of P2, T1, T4 and M2. Time subscripts are dropped; the period is static.

3.1 Prices

Write (2.5) and $mc_0 = w^h$ in vector form. Let $\log A$ be the vector with entries $\varepsilon_i + a_i(\Omega_i)$ for $i \geq 1$ and 0 for $i = 0$, and let $\alpha \equiv e_0$. Then

$$p = \Omega p + \alpha w^h - \log A.$$

Row 0 of Ω is zero (T4), so row 0 reads $p_0 = w = w^h$. The Leontief inverse $L \equiv (I - \Omega)^{-1}$ exists because Ω is substochastic with a zero row and every sector reaches sector 0 through a chain of positive shares. Hence

$$p = L\alpha\, w^h - L \log A. \tag{3.1}$$

Because each row of Ω sums to $1 - \alpha_i$ (T1, T4), $(I - \Omega)\mathbf{1} = \alpha$, hence $L\alpha = \mathbf{1}$, and (3.1) becomes

$$p_i = w - \big[L \log A\big]_i \qquad \text{for all } i. \tag{3.2}$$

Every price equals the wage minus the cumulated productivity of the sector's own supply chain. Prices in wage units are linear in $\log A$ and therefore, given E3, Gaussian. This is KMNT Lemma 1 with the union as sector 0.

3.2 Sales, Domar weights and GDP

Sales of sector i equal final demand plus intermediate demand from all sectors. Using (2.1), the factor demands in 2.2(a), and $MC_j = P_j$:

$$P_i Y_i = \beta_i P^C C + \sum_j \Omega_{ji} P_j Y_j.$$

In vector form with $s_i \equiv P_i Y_i$: $s = \beta\, P^C C + \Omega^\top s$, so $s^\top = \beta^\top L\, P^C C$. Define the **Domar weights**

$$\lambda^\top \equiv \beta^\top L = \beta^\top (I - \Omega)^{-1}, \qquad P_i Y_i = \lambda_i\, P^C C. \tag{3.3}$$

For the union, $\lambda_0 = \beta^\top L e_0$. Post-multiplying $L\alpha = \mathbf{1}$ by $\beta^\top$ gives $\lambda^\top \alpha = \lambda_0 = 1$: union sales equal nominal GDP, $WL = P^C C$. Nominal GDP is the wage bill; profits net of the subsidy are zero because $P_i = MC_i$ and the subsidy is rebated (M2). Hence

$$p^C = \beta^\top p = w - \lambda^\top \log A, \qquad c = w + l - p^C = l + \lambda^\top \log A. \tag{3.4}$$

Define aggregate log TFP

$$z \equiv \lambda^\top \log A = \sum_{i=1}^{N} \lambda_i \big(\varepsilon_i + a_i(\Omega_i)\big). \tag{3.5}$$

Then $c = l + z$ exactly: real GDP is aggregate TFP times hours. This is Hulten's theorem holding exactly (not just to first order) because all shares are constant.

3.3 Hours, output and the natural allocation

Combine (3.4) with the labor supply condition (2.2), $w^h - p^C = \gamma c + \phi l$, and $w = w^h$: the real wage is $w - p^C = z$, so $z = \gamma c + \phi l = \gamma(l + z) + \phi l$, giving

$$l^{nat} = \frac{1 - \gamma}{\gamma + \phi}\, z, \qquad c^{nat} = y^{nat} = \frac{1 + \phi}{\gamma + \phi}\, z. \tag{3.6}$$

These are R23 Lemma 1 with $\log A$ replaced by the exact TFP vector. Under P3 ($\gamma > 1$) hours fall when productivity rises: the income effect dominates. The steady state of the regime is (3.2)–(3.6) evaluated at $\log A = \mu + a(\Omega^*)$ (with $\varepsilon_0 = a_0 = 0$), and the sticky-price block linearizes around it (N2).

4. The network block

4.1 What a producer maximizes

Under I1–I2, producer f in sector i evaluates a candidate technique Ω_i' by the expected discounted profit it would earn if it used Ω_i' while resetting its price every period, in an economy where all other producers use the equilibrium network Ω^*. Two facts simplify this.

First, a producer with pre-subsidy marginal cost MC' and a flexible price charges the markup $\mathcal{M}$ over its effective marginal cost $MC'/\mathcal{M}$, so $P_f = MC'$. Its profit per period is

$$\Pi_f = \left(P_f - \frac{MC'}{\mathcal{M}}\right)\left(\frac{P_f}{P_i}\right)^{-\theta} Y_i = \left(1 - \frac{1}{\mathcal{M}}\right) MC' \left(\frac{MC'}{P_i}\right)^{-\theta} Y_i = \frac{1}{\theta}\left(\frac{MC'}{P_i}\right)^{1-\theta} P_i Y_i. \tag{4.1}$$

Second, by (3.3), $P_i Y_i = \lambda_i P^C C$ and the state price is $\Lambda_t = \rho^t C_t^{-\gamma} / P_t^C$, so

$$\Lambda_t P_{it} Y_{it} = \rho^t \lambda_i C_t^{1-\gamma}. \tag{4.2}$$

The nominal-GDP scaling and the price level cancel: what remains is real consumption raised to $1 - \gamma$. Therefore the value of the technique is

$$\mathcal{J}_i(\Omega_i') = \frac{\lambda_i}{\theta} \sum_t \rho^t\, \mathbb{E}\left[C_t^{1-\gamma} \left(\frac{MC_{it}(\Omega_i')}{P_{it}}\right)^{1-\theta}\right]. \tag{4.3}$$

Under I1 every period has the same unconditional distribution, so the ρ^t sum is a constant multiple of a single-period expectation and can be dropped. Write the bracket in logs. From (2.5) and (3.6),

$$\log\left[C^{1-\gamma}\left(\frac{MC_i(\Omega_i')}{P_i}\right)^{1-\theta}\right] = (1-\gamma)c + (1-\theta)\Big[\Omega_i'^\top p - \varepsilon_i - a_i(\Omega_i') - p_i\Big],$$

which is Gaussian because c and p are linear in ε (Section 3). Define the log **effective discount factor**

$$\tilde{\lambda} \equiv (1 - \gamma)\, c^{nat} = -\Gamma z, \qquad \Gamma \equiv \frac{(\gamma - 1)(1 + \phi)}{\gamma + \phi}, \tag{4.4}$$

using (3.6). Using $\mathbb{E}[e^X] = \exp(\mathbb{E}X + \frac{1}{2}\mathbb{V}X)$ for Gaussian X, the log of the expectation in (4.3) is

$$\log \mathcal{J}_i(\Omega_i') = \text{const} + (1-\theta)\Big[\Omega_i'^{\top}\mathbb{E}p - a_i(\Omega_i')\Big] + \tfrac{1}{2}\mathbb{V}\Big[\tilde{\lambda} + (1-\theta)\big(\Omega_i'^{\top}p - \varepsilon_i - a_i(\Omega_i') - p_i\big)\Big], \tag{4.5}$$

where the constant collects terms that do not depend on Ω_i', including $\mathbb{E}[\tilde{\lambda} - (1-\theta)(\varepsilon_i + p_i)]$.

4.2 First-order condition and the risk-adjusted price

Differentiate (4.5) with respect to Ω_{ik}'. The deterministic part gives $(1-\theta)(\mathbb{E}p_k - \partial a_i/\partial\Omega_{ik})$. The variance part gives $\frac{1}{2}\cdot 2\operatorname{Cov}\big[\tilde{\lambda} + (1-\theta)(\Omega_i'^{\top}p - \varepsilon_i - a_i - p_i),\,(1-\theta)p_k\big]$, since only p_k is stochastic in $\partial(\cdot)/\partial\Omega_{ik}'$. Collecting and dividing through by $(1-\theta) \neq 0$, with multiplier ν_i on $\mathbf{1}^{\top}\Omega_i' = 1$:

$$\mathbb{E}p_k - \frac{\partial a_i}{\partial \Omega_{ik}} + \operatorname{Cov}[\tilde{\lambda}, p_k] + (1-\theta)\operatorname{Cov}\big[\Omega_i'^{\top}p - \varepsilon_i - a_i(\Omega_i') - p_i,\ p_k\big] - \nu_i = 0. \tag{4.6}$$

Now impose symmetry: in equilibrium $\Omega_i' = \Omega_i^*$, so $\Omega_i'^{\top}p - \varepsilon_i - a_i(\Omega_i') = mc_i = p_i$ by (2.5) and Section 3. The last covariance in (4.6) is therefore $\operatorname{Cov}[p_i - p_i, p_k] = 0$ **at the equilibrium technique**, and the condition collapses to

$$\frac{\partial a_i(\Omega_i^*)}{\partial \Omega_{ik}} = R_k - \nu_i \quad \text{for all } k, \qquad R \equiv \mathbb{E}[p] + \operatorname{Cov}[p, \tilde{\lambda}]. \tag{4.7}$$

R is the vector of **risk-adjusted prices**: the expected price plus its covariance with the household's effective marginal valuation. This is KMNT Lemma 2 with two changes. (i) The producer is a monopolist, not a price-taker; the CES structure adds the last term of (4.6), which vanishes only at the symmetric equilibrium, so (4.7) is a statement about the equilibrium first-order condition, not about the objective. (ii) The discount-factor weight is Γ instead of $\gamma - 1$, because elastic labor scales sales with $C^{1-\gamma}$ rather than with $C^{-\gamma}$ alone (equation (4.2)).

Using (3.2) with $w = 0$ (wage numeraire) and (3.5),

$$\mathbb{E}p = -L(\mu + a(\Omega^*)), \qquad \operatorname{Cov}[p, \tilde{\lambda}] = -\Gamma \operatorname{Cov}[-L\log A,\ \lambda^{\top}\log A] = \Gamma\, L\Sigma L^{\top}\beta, \tag{4.8}$$

where Σ is padded with a zero row and column for sector 0 and $\lambda = L^{\top}\beta$. So $R_0 = 0$ and, for $k \geq 1$, R_k is the expected price of k in wage units plus Γ times the covariance of k's price with aggregate TFP. A supplier whose price rises when the economy is poor ($\operatorname{Cov}[p_k, z] < 0$, i.e. $[L\Sigma\lambda]_k > 0$) is penalized in proportion to Γ.

4.3 Second-order condition

The Hessian of (4.5) with respect to Ω_i' is $(1-\theta)(-H_i) + (1-\theta)^2\Sigma_p$, where $\Sigma_p \equiv \mathbb{V}[p] = L\Sigma L^{\top}$. For a maximum it must be negative definite on the tangent space of the simplex; dividing by $\theta - 1 > 0$,

$$H_i + (\theta - 1)\,\Sigma_p \prec 0 \quad \text{on } \{v : \mathbf{1}^{\top}v = 0\}. \tag{4.9}$$

The second term is positive semidefinite: a monopolist's profit (4.1) is convex in its own cost, so cost variance *raises* expected profit with strength $\theta - 1$. Condition (4.9) says the curvature of the TFP shifter must beat this convexity. KMNT's price-taking producer has objective Hessian $-H_i + \Sigma_p \succ 0$ for a minimum and needs no such condition.

4.4 Closed form under T2

With quadratic a_i, $\partial a_i/\partial\Omega_i = H_i(\Omega_i - \Omega_i^{\circ})$ and (4.7) reads $H_i(\Omega_i^* - \Omega_i^{\circ}) = R - \nu_i\mathbf{1}$, so $\Omega_i^* = \Omega_i^{\circ} + H_i^{-1}(R - \nu_i\mathbf{1})$. Imposing $\mathbf{1}^{\top}\Omega_i^* = 1 = \mathbf{1}^{\top}\Omega_i^{\circ}$ gives $\mathbf{1}^{\top}H_i^{-1}(R - \nu_i\mathbf{1}) = 0$, so $\nu_i = \mathbf{1}^{\top}H_i^{-1}R/\mathbf{1}^{\top}H_i^{-1}\mathbf{1}$ and

$$\Omega_i^* = \Omega_i^{\circ} + \hat{H}_i^{-1}R, \qquad \hat{H}_i^{-1} \equiv H_i^{-1} - \frac{H_i^{-1}\mathbf{1}\mathbf{1}^{\top}H_i^{-1}}{\mathbf{1}^{\top}H_i^{-1}\mathbf{1}}. \tag{4.10}$$

$\hat{H}_i^{-1}$ annihilates $\mathbf{1}$, so only *relative* risk-adjusted prices move the technique; adding a constant to R changes nothing, as it should under a numeraire choice. The equilibrium network is the fixed point $\Omega^* = \Omega^{\circ} + \hat{H}^{-1}R(\Omega^*)$ with $R(\Omega^*)$ from (4.8).

4.5 Planner equivalence and the risk weight

The flex-price equilibrium is undistorted (M2, M3), so by the first welfare theorem the equilibrium network maximizes expected household utility. Using $C = Z^{(1+\phi)/(\gamma+\phi)}$ and $L = Z^{(1-\gamma)/(\gamma+\phi)}$ from (3.6), indirect utility as a function of $Z = e^z$ is

$$V(Z) = \frac{Z^k}{1-\gamma} - \frac{Z^k}{1+\phi} = \frac{Z^k}{k}, \qquad k \equiv \frac{(1-\gamma)(1+\phi)}{\gamma+\phi} = -\Gamma,$$

since $\frac{1}{1-\gamma} - \frac{1}{1+\phi} = \frac{\gamma+\phi}{(1-\gamma)(1+\phi)}$. With z Gaussian, $\mathbb{E}V = \frac{1}{k}\exp(k\mathbb{E}z + \frac{1}{2}k^2\mathbb{V}z)$. For $\gamma > 1$, $k < 0$ and $1/k < 0$, so maximizing $\mathbb{E}V$ is minimizing $k\mathbb{E}z + \frac{1}{2}k^2\mathbb{V}z$, i.e. (dividing by $k < 0$) maximizing

$$\mathcal{W}(\Omega) = \mathbb{E}[z] - \tfrac{1}{2}\,\Gamma\,\mathbb{V}[z] = \lambda^\top\big(\mu + a(\Omega)\big) - \tfrac{1}{2}\,\Gamma\,\lambda^\top\Sigma\lambda. \tag{4.11}$$

The same expression results for $\gamma < 1$ (then $k > 0$, $1/k > 0$, and $\Gamma < 0$: the planner rewards variance) and for $\gamma = 1$ by the log limit ($\Gamma = 0$). This is KMNT's planner objective with $\rho_{KMNT} - 1$ replaced by Γ; as $\phi \to \infty$, $\Gamma \to \gamma - 1$. KMNT Prop. 1 shows the fixed point of (4.7)–(4.8) is the maximizer of (4.11); the proof is unchanged by the substitution, and the derivation in 4.1–4.2 verifies its key step — that producers discount with weight Γ — directly from the decentralized problem.

4.6 Comparative statics (inherited)

Everything in KMNT Sections 5–7 applies with $\rho - 1 \to \Gamma$: Prop. 2 (own Domar weight falls in own variance), Prop. 3 ($d\lambda/d\gamma = -H^{-1}\partial\mathcal{E}/\partial\gamma$ for any belief parameter $\gamma \in \{\mu_i, \Sigma_{ij}\}$, with $\mathcal{E} \equiv \mu - \Gamma\Sigma\lambda$), Cor. 3 (sign of cross-effects given by global complementarity), and Lemma 9 (with $H_i = -\bar{\kappa}\hat{H}_i$, $\Omega_i^* = \Omega_i^\circ + \bar{\kappa}^{-1}(\hat{H}_i^\circ)^{-1}R^\circ + O(\bar{\kappa}^{-2})$).

5. The sticky-price block

From here on all variables are log-deviations from the regime's steady state (N2), so $\log A_t \equiv \varepsilon_t - \mu$ with $\log A_{0t} = 0$, and $a_i(\Omega_i^*)$ is absorbed into the steady state. $\Omega \equiv \Omega^*$, $L \equiv (I - \Omega^*)^{-1}$, $\lambda \equiv \lambda^*$ throughout.

5.1 From the Calvo problem to the sectoral Phillips curve

Reset price. Write $x_{t,t+s} \equiv \Lambda_{t,t+s}P_{i,t+s}^{\theta}Y_{i,t+s}$ for the weight in (2.6). In the zero-inflation steady state $P^* = P = MC$ and $x_{t,t+s} = \rho^s\bar{x}$. Because the bracket $[P_{it}^* - MC_{i,t+s}]$ is zero in steady state, first-order deviations of $x_{t,t+s}$ multiply zero and drop out; only the bracket's own deviation survives. Dividing (2.6) by $\bar{x}\bar{P}$,

$$\sum_{s=0}^{\infty}(1-\delta_i)^s\rho^s\,\mathbb{E}_t\big[p_{it}^* - mc_{i,t+s}\big] = 0 \implies p_{it}^* = \big(1-\rho(1-\delta_i)\big)\sum_{s=0}^{\infty}\big(\rho(1-\delta_i)\big)^s\mathbb{E}_t\,mc_{i,t+s}, \tag{5.1}$$

using $\sum_s(\rho(1-\delta_i))^s = 1/(1-\rho(1-\delta_i))$. Equivalently, splitting off $s = 0$,

$$p_{it}^* = \big(1-\rho(1-\delta_i)\big)mc_{it} + \rho(1-\delta_i)\,\mathbb{E}_t p_{i,t+1}^*. \tag{5.2}$$

Price index. Log-linearizing (2.7): $(1-\theta)p_{it} = (1-\delta_i)(1-\theta)p_{i,t-1} + \delta_i(1-\theta)p_{it}^*$, so

$$p_{it} = (1-\delta_i)p_{i,t-1} + \delta_i p_{it}^* \quad\Longleftrightarrow\quad \pi_{it} = \delta_i\big(p_{it}^* - p_{i,t-1}\big). \tag{5.3}$$

Combining. Subtract $p_{i,t-1}$ from both sides of (5.2):

$$p_{it}^* - p_{i,t-1} = \big(1-\rho(1-\delta_i)\big)\big(mc_{it} - p_{i,t-1}\big) + \rho(1-\delta_i)\big(\mathbb{E}_t p_{i,t+1}^* - p_{i,t-1}\big).$$

Use $mc_{it} - p_{i,t-1} = (mc_{it} - p_{it}) + \pi_{it}$ and, from (5.3) shifted forward, $\mathbb{E}_t p_{i,t+1}^* - p_{i,t-1} = (\mathbb{E}_t p_{i,t+1}^* - p_{it}) + \pi_{it} = \mathbb{E}_t\pi_{i,t+1}/\delta_i + \pi_{it}$. Then multiply through by δ_i and use (5.3) on the left:

$$\pi_{it} = \delta_i\big(1-\rho(1-\delta_i)\big)\big(mc_{it} - p_{it}\big) + \delta_i\big[(1-\rho(1-\delta_i)) + \rho(1-\delta_i)\big]\pi_{it} + \rho(1-\delta_i)\mathbb{E}_t\pi_{i,t+1}.$$

The middle bracket equals 1, so $\pi_{it}(1-\delta_i) = \delta_i(1-\rho(1-\delta_i))(mc_{it}-p_{it}) + \rho(1-\delta_i)\mathbb{E}_t\pi_{i,t+1}$. Dividing by $1-\delta_i$:

$$\boxed{\pi_{it} = \rho\,\mathbb{E}_t\pi_{i,t+1} + \kappa_i\big(mc_{it} - p_{it}\big), \qquad \kappa_i \equiv \frac{\delta_i\big(1-\rho(1-\delta_i)\big)}{1-\delta_i}} \tag{5.4}$$

This holds for every sector $i \in \{0,\dots,N\}$, with mc_{0t} given by (2.8). In matrix form, $K \equiv \mathrm{diag}(\kappa_0,\dots,\kappa_N)$:

$$\pi_t = \rho\,\mathbb{E}_t\pi_{t+1} + K\big(mc_t - p_t\big). \tag{5.5}$$

5.2 Marginal-cost gaps in terms of prices, productivity and the output gap

Stack (2.5) for $i \geq 1$ and (2.8) for $i = 0$:

$$mc_t = \Omega p_t + \alpha\, w_t^h - \log A_t, \qquad w_t^h = p_t^C + \gamma c_t + \phi l_t. \tag{5.6}$$

Three first-order facts convert w_t^h into observables. (a) Goods-market clearing with no investment: $c_t = y_t$. (b) Definition of the gap: $y_t = y_t^{nat} + \tilde{y}_t$. (c) *Output gap equals hours gap*: aggregate output is $y_t = \lambda^\top \log A_t + l_t + O(\|\cdot\|^2)$ under sticky prices as well as flexible ones, because the steady state is efficient (M2) and by the envelope theorem the misallocation of inputs across sectors caused by relative-price distortions has no first-order effect on aggregate output (R23 Remark 5). Subtracting the natural counterpart, $\tilde{y}_t = \tilde{l}_t$. Therefore

$$\gamma c_t + \phi l_t = \gamma\big(y_t^{nat} + \tilde{y}_t\big) + \phi\big(l_t^{nat} + \tilde{y}_t\big) = \frac{\gamma(1+\phi)+\phi(1-\gamma)}{\gamma+\phi}\lambda^\top \log A_t + (\gamma+\phi)\tilde{y}_t = \lambda^\top \log A_t + (\gamma+\phi)\tilde{y}_t,$$

using (3.6) and $\gamma(1+\phi)+\phi(1-\gamma) = \gamma+\phi$. So

$$w_t^h = \beta^\top p_t + \lambda^\top \log A_t + (\gamma+\phi)\tilde{y}_t. \tag{5.7}$$

The natural real wage rises one-for-one with aggregate TFP; the gap adds $(\gamma+\phi)\tilde{y}_t$. Substituting (5.7) into (5.6) and subtracting p_t:

$$mc_t - p_t = -\big[(I-\Omega) - \alpha\beta^\top\big]p_t - \big[I - \alpha\lambda^\top\big]\log A_t + \alpha(\gamma+\phi)\tilde{y}_t. \tag{5.8}$$

Two identities collapse the two brackets into one operator. From $(I-\Omega)\mathbf{1} = \alpha$:

$$(I-\Omega) - \alpha\beta^\top = (I-\Omega)\big(I - \mathbf{1}\beta^\top\big) \equiv Q,$$

and from $\lambda^\top = \beta^\top L$, $(I-\Omega)L = I$:

$$I - \alpha\lambda^\top = (I-\Omega)L - (I-\Omega)\mathbf{1}\beta^\top L = (I-\Omega)\big(I - \mathbf{1}\beta^\top\big)L = QL.$$

Hence

$$\boxed{mc_t - p_t = -\,Q\big(p_t + L\log A_t\big) + \alpha(\gamma+\phi)\tilde{y}_t} \tag{5.9}$$

Interpretation. By (3.2) the natural price vector is $p_t^{nat} = \mathbf{1}w_t - L\log A_t$, and $Q\mathbf{1} = (I-\Omega)(\mathbf{1}-\mathbf{1}) = 0$, so $Q(p_t + L\log A_t) = Q(p_t - p_t^{nat})$: the marginal-cost gap of sector i is minus the gap between actual and natural *relative* prices, weighted by i's input shares, plus (for the union only) the labor-cost gap $(\gamma+\phi)\tilde{y}_t$. Following R23, define the state

$$\chi_t \equiv p_{t-1} + L\log A_t, \qquad p_t + L\log A_t = \chi_t + \pi_t. \tag{5.10}$$

5.3 The sectoral Phillips-curve system and its map to R23

Substitute (5.9)–(5.10) into (5.5):

$$\pi_t = \rho\,\mathbb{E}_t\pi_{t+1} + K\alpha(\gamma+\phi)\tilde{y}_t - KQ\big(\chi_t + \pi_t\big) \implies (I+KQ)\pi_t = \rho\,\mathbb{E}_t\pi_{t+1} + K\alpha(\gamma+\phi)\tilde{y}_t - KQ\chi_t.$$

Let $M \equiv (I+KQ)^{-1}$ and define

$$V \equiv MKQ = I - M, \qquad B \equiv MK\alpha. \tag{5.11}$$

($V = I - M$ because $I - (I + KQ)^{-1} = (I + KQ)^{-1}(I + KQ - I) = (I + KQ)^{-1}KQ$.) Then

$$\boxed{\pi_t = \rho\,(I - V)\,\mathbb{E}_t\pi_{t+1} + B(\gamma + \phi)\tilde{y}_t - V\chi_t, \qquad p_t = p_{t-1} + \pi_t} \tag{5.12}$$

This is R23 Proposition 2. The state vector is $(p_{t-1}, \log A_t)$ with $\log A_t = \eta \log A_{t-1} + \xi_t$.

Map to R23's rigidity matrix. R23 writes V and B in terms of $\Delta = \mathrm{diag}(\hat{\delta}_i)$. Define $\hat{\delta}_i \equiv \kappa_i/(1+\kappa_i)$, so $K = \Delta(I - \Delta)^{-1}$. Explicitly,

$$\hat{\delta}_i = \frac{\delta_i(1 - \rho(1 - \delta_i))}{(1 - \delta_i) + \delta_i(1 - \rho(1 - \delta_i))} = \frac{\delta_i\big(1 - \rho(1 - \delta_i)\big)}{1 - \rho\,\delta_i(1 - \delta_i)}, \tag{5.13}$$

which is R23's $\hat{\delta}_i(\delta_i, \rho)$; $\hat{\delta}_i \in (0,1)$, $\hat{\delta}_i \to 1$ as $\delta_i \to 1$ and $\hat{\delta}_i \to 0$ as $\delta_i \to 0$. Now

$$I + KQ = (I - \Delta)^{-1}\big[(I - \Delta) + \Delta Q\big] = (I - \Delta)^{-1}\big[I - \Delta + \Delta(I - \Omega) - \Delta\alpha\beta^\top\big] = (I - \Delta)^{-1}\big[I - \Delta\Omega - \Delta\alpha\beta^\top\big],$$

so $M = \big[I - \Delta\Omega - \Delta\alpha\beta^\top\big]^{-1}(I - \Delta)$. Apply the Sherman–Morrison identity $[G - uv^\top]^{-1} = G^{-1} + G^{-1}uv^\top G^{-1}/(1 - v^\top G^{-1}u)$ with $G = I - \Delta\Omega$, $u = \Delta\alpha$, $v = \beta$, and define

$$\delta^\beta \equiv \beta^\top(I - \Delta\Omega)^{-1}\Delta \;\; \text{(row vector)}, \qquad \delta^\beta\alpha \equiv \beta^\top(I - \Delta\Omega)^{-1}\Delta\alpha \;\; \text{(scalar)}, \qquad B \equiv \frac{(I - \Delta\Omega)^{-1}\Delta\alpha}{1 - \delta^\beta\alpha}. \tag{5.14}$$

Then $[I - \Delta\Omega - \Delta\alpha\beta^\top]^{-1} = (I - \Delta\Omega)^{-1} + B\beta^\top(I - \Delta\Omega)^{-1}$, and:

The gap coefficient. $MK\alpha = [I - \Delta\Omega - \Delta\alpha\beta^\top]^{-1}(I - \Delta)\Delta(I - \Delta)^{-1}\alpha = [(I - \Delta\Omega)^{-1} + B\beta^\top(I - \Delta\Omega)^{-1}]\Delta\alpha = (I - \Delta\Omega)^{-1}\Delta\alpha\Big(1 + \frac{\delta^\beta\alpha}{1 - \delta^\beta\alpha}\Big) = B$, confirming that the B of (5.11) is the B of (5.14).

The cost-push matrix. $V = MKQ = [(I - \Delta\Omega)^{-1} + B\beta^\top(I - \Delta\Omega)^{-1}]\Delta\big[(I - \Omega) - \alpha\beta^\top\big]$. Expanding the four products,

$$V = (I - \Delta\Omega)^{-1}\Delta(I - \Omega) - (I - \Delta\Omega)^{-1}\Delta\alpha\,\beta^\top + B\,\delta^\beta(I - \Omega) - B\,(\delta^\beta\alpha)\,\beta^\top.$$

The second term is $-B(1 - \delta^\beta\alpha)\beta^\top$ by (5.14); adding the fourth gives $-B\beta^\top = -B\lambda^\top(I - \Omega)$ since $\beta^\top = \lambda^\top(I - \Omega)$. Therefore

$$V = \Big[(I - \Delta\Omega)^{-1}\Delta - B\big(\lambda^\top - \delta^\beta\big)\Big](I - \Omega), \tag{5.15}$$

which is R23's expression. The operator $(I - \Delta\Omega)^{-1}\Delta$ is the rigidity-discounted Leontief inverse: entry (i,j) sums over all supply-chain paths from i back to j, each step discounted by the reset probability of the sector at that step.

5.4 Closing the output gap (G1)

Under G1 the gap term in (5.12) is identically zero:

$$\pi_t = \rho\,(I - V)\,\mathbb{E}_t\pi_{t+1} - V\chi_t. \tag{5.16}$$

R23 Lemma 4 establishes that, with $\delta_i < 1$ for all i (N1) and AR(1) productivity (E3), (5.16) together with (5.10) has a unique bounded solution, and that it is implemented by a Taylor rule on the divine-coincidence index. All network effects on inflation under G1 run through V and χ_t; the slope B is switched off by construction.

5.5 Inflation indices, slopes and cost-push residuals

For any weight vector φ with $\varphi^\top\mathbf{1} = 1$, define $\pi_t^\varphi \equiv \varphi^\top\pi_t$ and $\delta^\varphi \equiv \varphi^\top(I - \Delta\Omega)^{-1}\Delta$. Pre-multiplying (5.12) by $\varphi^\top$:

$$\pi_t^\varphi = \rho\,\varphi^\top(I - V)\mathbb{E}_t\pi_{t+1} + \kappa^\varphi(\gamma + \phi)\tilde{y}_t + u_t^\varphi, \tag{5.17}$$

with

$$\kappa^\varphi \equiv \varphi^\top B = \frac{\delta^\varphi\alpha}{1 - \delta^\beta\alpha}, \qquad u_t^\varphi \equiv -\varphi^\top V\chi_t = \Big[\kappa^\varphi\big(\lambda^\top - \delta^\beta\big) - \delta^\varphi\Big](I - \Omega)\chi_t, \tag{5.18}$$

the second by pre-multiplying (5.15) by $-\varphi^\top$. This is R23 Corollary 2. The expectation term is not proportional to $\mathbb{E}_t \pi^{\varphi}_{t+1}$ in general, so (5.17) is not a closed aggregate Phillips curve.

For the CPI ($\varphi = \beta$), $\kappa^C = \delta^\beta\alpha/(1-\delta^\beta\alpha)$ and the bracket in (5.18) simplifies:

$$\frac{\delta^\beta\alpha(\lambda^\top - \delta^\beta)}{1-\delta^\beta\alpha} - \delta^\beta = \frac{\delta^\beta\alpha\,\lambda^\top - \delta^\beta\alpha\,\delta^\beta - \delta^\beta + \delta^\beta\alpha\,\delta^\beta}{1-\delta^\beta\alpha} = \frac{\delta^\beta\alpha\,\lambda^\top - \delta^\beta}{1-\delta^\beta\alpha},$$

so

$$u^C_t = \frac{\delta^\beta\alpha\,\lambda^\top - \delta^\beta}{1-\delta^\beta\alpha}\,(I-\Omega)\chi_t. \tag{5.19}$$

On impact of a shock with $p_{t-1} = 0$, $(I-\Omega)\chi_t = \log A_t$, so $u^C_t = v^C \log A_t$ with the **exposure vector**

$$v^C \equiv \frac{\delta^\beta\alpha\,\lambda^\top - \delta^\beta}{1-\delta^\beta\alpha}. \tag{5.20}$$

Entry j is the impact cost-push in CPI from a unit productivity shock in j: the wage channel $\delta^\beta\alpha\,\lambda_j$ (wages fall with aggregate TFP, discounted by wage and price rigidity along every path) minus the direct channel δ^β_j (the rigidity-discounted exposure of consumption to j's price), scaled by the wage–price multiplier $1/(1-\delta^\beta\alpha)$. Under G1, $\pi^C_t = \rho\beta^\top(I-V)\mathbb{E}_t\pi_{t+1} + u^C_t$: the whole CPI response is the cost-push term and its expected continuation.

5.6 The IS curve (needed only when G1 is relaxed)

Log-linearizing (2.3) with $c_t = y_t$: $y_t = \mathbb{E}_t y_{t+1} - \frac{1}{\gamma}\big(i_t - \mathbb{E}_t\pi^C_{t+1} + \log\rho\big)$. Subtracting the natural counterpart and defining the natural rate $r^n_t \equiv -\log\rho + \gamma\,\mathbb{E}_t\big(y^{nat}_{t+1} - y^{nat}_t\big)$,

$$\tilde{y}_t = \mathbb{E}_t\tilde{y}_{t+1} - \frac{1}{\gamma}\big(i_t - \mathbb{E}_t\pi^C_{t+1} - r^n_t\big), \qquad r^n_t = -\log\rho + \gamma(\eta-1)\frac{1+\phi}{\gamma+\phi}\lambda^\top \log A_t. \tag{5.21}$$

Note that r^n_t loads on $\log A_t$ through $\lambda = \lambda^*$: a regime change moves the natural rate's sensitivity to every sectoral shock, a third channel (beyond exposure and slope) through which the network affects dynamics once the gap is not closed.

6. (Main) Issues

- **Frequency separation: producers evaluate the technique choice problem at flex-price objects that they do not face in reality.** This assumption guarantees a closed form and allows most of the network-block theorems to hold as in KMNT paper. However, this omits a channel of technique choice: producers are pushed toward sticky-price suppliers. Under M4 price stickiness is the same between foreign and domestic, so this issue does not affect reshoring/delocalization decisions.
- **Foreign producers buy labor from the labor union just as all the other domestic sectors.** This keeps the input-output matrix invertible and thus guarantees the existence of the Leontief inverse. Need a way to incorporate imperfect labor mobility. Can changing the model of wage rigidity (dropping union sector in Ω_i) be a solution?

Each item states the issue, the assumption(s) responsible, and the resolution or the flag to carry.

1. The inflationary sign of a foreign supply shock depends on wage rigidity, not on the network. From (5.20), in the one-sector economy (union 0, final good C, $\Omega_{C0} = 1$) one finds $v_C^C = -\hat{\delta}_C(1-\hat{\delta}_L)/(1-\hat{\delta}_C\hat{\delta}_L)$: a negative productivity shock raises CPI cost-push only if $\hat{\delta}_L < 1$. With flexible wages ($\hat{\delta}_L = 1$) the residual is zero — the divine coincidence — and in multi-sector economies with flexible wages an upstream negative shock is *deflationary* (R23 Example 2). *Source:* N1 with $\delta_L < 1$. *Status:* intended; N1 must not be relaxed to flexible wages. We know this from macro 101. In the single sector economy without wage rigidity we have divine coincidence

2. The technique choice of a monopolistic producer has a second-order condition that KMNT's price-taker does not. Equation (4.9): $H_i + (\theta-1)\Sigma_p \prec 0$ on the simplex tangent space. A CES monopolist's profit is convex in own cost (4.1), so it values cost variance with weight $\theta - 1$; if the TFP shifter is not curved enough the producer prefers a corner technique to gamble on cost. *Source:* M1 (CES, monopolistic competition, needed for Calvo) combined with T2 (finite curvature H_i). *Resolution:* impose $\bar{\kappa}$ large (A5 of the design document), which makes $-H_i$ dominate; verify (4.9) numerically at the calibrated Σ_p. The first-order condition (4.7) is unaffected, so all KMNT comparative statics still apply. Mention to MW because it is a difference compared to the KMNT paper. The assumption of large $\bar{\kappa}$ makes sense and is used in KMNT, as well.

3. Risk aversion γ in the network block requires $\gamma > 1$, which makes hours countercyclical to productivity. $\Gamma > 0$ iff $\gamma > 1$ (P3); then (3.6) gives $l^{nat} = \frac{1-\gamma}{\gamma+\phi} z < 0$ for $z > 0$. *Source:* P1, P3, E1 (no capital to absorb the income effect). *Status:* a known property of the textbook NK model with $\gamma > 1$; acceptable for the questions asked, but note that the same parameter drives both the flight-to-safety motive and the hours response. Also this assumption of $\gamma > 1$ makes sense and is used in KMNT, as well. See footnote 10 in paper w/ supplement.

4. Frequency separation evaluates the technique at flex-price objects that the producer does not actually face. Under G1, $c_t = c_t^{nat}$ to first order, so $\tilde{\lambda}$ in (4.4) is correct to first order; but sectoral prices and quantities deviate from their natural values at first order, and a Calvo producer stuck at an old price bears input-cost variance directly in its markup — a hedging motive that (4.7) omits. *Source:* I2. *Status:* the approximation error is small when shocks are persistent or the sourcing horizon is long (variance-ratio calculation in the design discussion); the omitted motive would push producers *toward* sticky-price suppliers, which under M4 is neutral between foreign and domestic. Flag as extension. MAIN DISCUSSION POINT WITH MW.

5. Techniques use unconditional moments although the current ε_0 may be known. (4.3) replaced $\sum_t \rho^t \mathbb{E}[\cdot]$ by a single unconditional expectation. With AR(1) shocks (E3) the conditional distribution at the start of the regime differs from the unconditional one. *Source:* I1. *Resolution:* interpret the choice as made behind the veil, or as the average over the regime's stationary distribution; either is exact if the regime is long relative to $1/(1-\eta)$.

6. A regime change moves the linearization point. N2 linearizes around the steady state of the regime, which has network $\Omega^*(\mu, \Sigma)$ and TFP levels $\mu + a(\Omega^*)$. After a fragmentation shock, Ω^* changes discretely, and inherited prices p_{t-1} were set under the old steady state. Computing the transition with the new (Ω^*, V) and $\chi_t = p_{t-1} + L \log A_t$ treats the difference between old and new natural prices as a one-off shock, which is valid to first order only if $\Delta\Omega$ is small. *Source:* N2, E4, T3. *Resolution:* keep $\bar{\kappa}$ large (then $\Delta\Omega = O(\bar{\kappa}^{-1})$, KMNT Lemma 9); otherwise compute the transition numerically as a change of steady state.

7. Ex-post shares are constant, so there is no price-induced reallocation. Under T1 a permanent rise in a foreign price does not change Ω_{if} ex post; all network movement is ex ante through R. *Source:* T1 (needed for exact log-linearity of prices and for (4.5) to be Gaussian). *Status:* a substantive restriction, stated openly; the robustness check is a CES nest between F and D in the ex-post technology, which leaves (5.12) unchanged to first order (cost shares are sufficient statistics by Shephard's lemma) but adds a direct price elasticity to Ω. Yes. A realized rise in a foreign price within a regime, however persistent, leaves Ω unchanged; Ω moves only when beliefs (μ, Σ) change. But my model is all about beliefs. I don't really care about within-regime network adjustment.

8. The log-normal mean effect. A rise in Σ_{ii} raises $\mathbb{E}[e^{\varepsilon_i}]$ by $\frac{1}{2}\Sigma_{ii}$; the objective (4.11) is in logs and does not see this, but level comparisons of GDP across regimes would. *Source:* E3. *Status:* KMNT App. H; ignore in the baseline and say so.

9. Correction to an earlier claim: the fully flexible limit is $0/0$, not zero. As $\Delta \to I$, $\delta^\beta \to \lambda^\top$ and $\delta^\beta \alpha \to 1$, so both numerator and denominator of (5.20) vanish. The limit depends on the direction of approach: with $\hat{\delta}_L = 1$ and any $\hat{\delta}_C$ in the one-sector economy, $v_C^C = 0$; with uniform $\hat{\delta}_i = 1 - \epsilon$, $\epsilon \to 0$, $v_C^C \to -\frac{1}{2}$ — a TFP shock is split evenly between price and wage adjustment however small the rigidity, because what matters is *relative* stickiness. The statement in `model_foundations.md` §7.2 that "$v^C \equiv 0$ under fully flexible prices" is wrong and should read: the CPI residual vanishes when wages are flexible and there is a single final sector (divine coincidence); in general it depends on the ratio of wage to price rigidity even in the flexible limit. *Source:* N1. *Status:* corrected below in the design document.

10. Sectors with $\delta_i = 1$ make κ_i infinite. (5.4) degenerates to $mc_{it} = p_{it}$, and K in (5.5) is unbounded; R23's invertibility caveat for $I - V$ is the same fact. *Source:* N1. *Status:* non-binding under M4 ($\delta_F = \delta_D < 1$); in the commodity variant, substitute the static condition for that sector rather than using κ_i.

11. The union's Phillips curve has the standard slope only because of pooling. Without P4, each labor type's marginal rate of substitution depends on its own hours, which depend on its relative wage, and the wage Phillips curve slope is $\kappa_0/(1 + \theta\phi)$ (Erceg–Henderson–Levin). *Source:* P4. *Status:* intended simplification; the union is "just a sector" only under P4. GTK, but the assumption is standard in the literature(?)

12. Subsidy financing and profits under sticky prices. In the flex economy the subsidy equals aggregate profit and $P^C C = WL$ exactly (3.4). Under sticky prices profits differ from the subsidy at first order, but the difference is a lump-sum transfer to a Ricardian household and does not affect first-order allocations. *Source:* M2, E1 (no distortionary taxes). *Status:* fine.

13. $y_t = c_t$ and $\tilde{y}_t = \tilde{l}_t$ are first-order statements under sticky prices. Price dispersion within sectors and relative-price misallocation across sectors are second order at an efficient steady state. *Source:* N2, M2. *Status:* fine; would fail if the steady state were distorted (no subsidy), in which case first-order terms from misallocation appear and R23's welfare block would be needed.

14. Closing the gap requires the central bank to know λ^*. After a regime change $y_t^{nat} = \frac{1+\phi}{\gamma+\phi}\lambda^{*\top} \log A_t$ loads on shocks through the new Domar weights; a bank using the old λ would misjudge the natural level. *Source:* I3, G1. *Status:* assumed; an informational-friction extension is possible but off the main line.

15. Foreign sectors buy domestic labor services. Under M4 a foreign supplier's marginal cost (2.5) contains the domestic wage p_{0t}, so domestic wage shocks transmit into "foreign" prices and foreign productivity shocks feed back into domestic labor demand exactly as domestic ones do. This is unrealistic for imports. *Source:* M4 (the price of keeping $L\alpha = \mathbf{1}$ and R23's lemmas intact). *Status:* the open-economy extension replaces the foreign sector's labor input with an exogenous factor; until then, "foreign" means "risky and productive", not "abroad".

16. Efficiency of foreign sourcing is parameterized twice. A foreign input can be attractive through a high μ_f (cheap) or a high ideal share Ω°_{if} (technologically preferred). Both enter R and Ω^* but only μ_f interacts with risk through $\mathcal{E} = \mu - \Gamma\Sigma\lambda$. *Source:* T2, E3. *Status:* choose one as the "efficiency side" of the trade-off and hold the other fixed, to keep comparative statics interpretable.

17. $\hat{\delta}_i$ is not the Calvo reset probability. By (5.13) a quarterly reset probability $\delta_i = 0.25$ implies $\hat{\delta}_i \approx 0.08$, and $\hat{\delta}_i = 0.25$ corresponds to $\delta_i \approx 0.44$. The illustrative calibrations used earlier in this project were stated in $\hat{\delta}$ units. *Source:* N1. *Status:* calibrate δ_i from price-duration data and convert with (5.13); do not set $\hat{\delta}$ directly.

18. Atomistic technique choice is consistent with one technique per sector. In (4.3) the producer takes λ_i, P_i, Y_i as given; its own Ω_{if} does not move them because it has measure zero (M1). Symmetry then delivers T3. No issue, but T3 is a consequence of M1 plus the uniqueness of the maximizer (which is where (4.9) matters), not an independent assumption.

7. Notation dictionary

Column "R23" gives Rubbo (2023)'s symbol for the same object; "KMNT" gives Kopytov et al. (2024)'s. A dash means the object does not appear in that paper. Where the meaning differs slightly the difference is noted.

Indices, time, sectors

Symbol	Meaning	R23	KMNT
t	period (quarter)	t	— (static)
$i, j, k \in \{0, \dots, N\}$	sectors; 0 = union	i, j; union = "sector I" in Remark 4	$i, j \in \{1, \dots, n\}$; labor is not a sector
$f \in [0, 1]$	producer / variety within a sector	f	— (representative firm)
N	number of goods sectors	N	n

Household

Symbol	Meaning	R23	KMNT
C_t, C_{it}	consumption aggregate, of good i	C_t, C_{it}	C, C_i
L_t	raw hours supplied	L_t	$L = 1$ (inelastic)
B_t, i_t	nominal bond, nominal rate	B_t, i_t	—
ρ, γ, ϕ	discount factor, risk aversion, inverse Frisch	ρ, γ, ϕ	ρ is risk aversion; no discounting; $\phi = \infty$
β	consumption shares	β	β
P_t^C, p_t^C	CPI, log CPI	P_t^C	P (consumption price index)
W_t^h, w_t^h	price of raw hours = nominal MRS	marginal cost of the union sector	—
$\Lambda_t, \Lambda_{t,t+s}$	nominal state price / SDF	Λ	Λ
$\tilde{\lambda}$	log effective discount weight $(1-\gamma)c^{nat} = -\Gamma z$	—	$\lambda \equiv \log \Lambda = (\rho - 1)\beta^\top p$
Π_t, T_t	profits, lump-sum tax	Π_t, T_t	—

Technology and productivity

Symbol	Meaning	R23	KMNT
Y_{ift}, Y_{it}	output of producer f, of sector i	Y_{ift}, Y_{it}	Q_i
X_{ifjt}	input j used by producer f in i; $j = 0$ is labor services	X_{ifjt}; labor L_{ift}	X_{ij}; labor L_i
Ω, Ω_i, Ω_{ij}	input-output matrix (incl. labor column 0), technique of i, share	Ω (goods only), α = labor shares	α, α_i, α_{ij}; labor share $1 - \sum_j \alpha_{ij}$
$\alpha \equiv e_0$	direct-labor-share vector (only the union)	α (general)	—
Ω_i°	ideal technique	—	α_i°
$a_i(\Omega_i)$	log TFP shifter, ≤ 0, quadratic	—	$a_i(\alpha_i) = \log A_i(\alpha_i)$

Symbol	Meaning	R23	KMNT
H_i, $\hat{H}_i^{-1}$	Hessian of a_i, its simplex-projected inverse	—	H_i, $(\hat{H}_i^{\circ})^{-1}$
$\bar{\kappa}$	scale of $-H_i$ in the large-curvature approximation	—	$\bar{\kappa}$
ζ_i	Cobb–Douglas normalization $\prod_j \Omega_{ij}^{-\Omega_{ij}}$	—	$\zeta(\alpha_i)$
$\varepsilon_t, \mu, \Sigma$	sectoral log productivity, its unconditional mean and covariance	—	ε, μ, Σ
$\log A_t \equiv \varepsilon_t - \mu$	productivity deviation	$\log A_t$ (with $A = 1$ in steady state)	—
η, ξ_t	AR(1) persistence, innovation	η_A	— (random walk in calibration)
A_{it}	exact TFP level $e^{\varepsilon_{it}+a_i}$	A_{it}	$e^{\varepsilon_i} A_i(\alpha_i)$
$L \equiv (I-\Omega)^{-1}$	Leontief inverse (incl. union)	$(I-\Omega)^{-1}$	$L(\alpha) = (I-\alpha)^{-1}$
$\lambda^\top \equiv \beta^\top L$	Domar weights ($\lambda_0 = 1$)	λ	ω
$z \equiv \lambda^\top \log A$, $Z = e^z$	aggregate log TFP	$\lambda^\top \log A_t$	y (log GDP, since $L = 1$)
$\Gamma \equiv \frac{(\gamma-1)(1+\phi)}{\gamma+\phi}$	planner/producer risk weight	—	$\rho - 1$
R	risk-adjusted prices $\mathbb{E}p + \mathrm{Cov}[p, \tilde{\lambda}]$	—	R
$\Sigma_p \equiv L\Sigma L^\top$	covariance of log prices	—	$L\Sigma L^\top$
$\mathcal{E} \equiv \mu - \Gamma\Sigma\lambda$	risk-adjusted TFP (comparative statics)	—	$\mathcal{E} = \mu - (\rho-1)\Sigma\omega$
$\mathcal{W}(\Omega)$	planner objective (4.11)	—	$\mathcal{W}$
ν_i	multiplier on $\mathbf{1}^\top\Omega_i = 1$	—	γ_i (on $\sum_j \alpha_{ij} \le \bar{\alpha}_i$)
τ^w	wasteful access wedge (folded into ε)	—	τ (App. L)

Market structure and nominal side

Symbol	Meaning	R23	KMNT
θ	within-sector CES elasticity	ε_i (sector-specific)	— (perfect competition)
$\mathcal{M} = \theta/(\theta-1)$	gross markup	μ	—
τ	input subsidy, $(1-\tau)\mathcal{M} = 1$	τ_i	— (App. K wedges)
P_{ift}, P_{it}, p_{it}	variety price, sectoral index, log	P_{ift}, P_{it}	P_i, p_i (in wage units)
$W_t \equiv P_{0t}$, w_t	wage paid by producers = union's price	W_t	$W = 1$ numeraire
MC_{it}, mc_{it}	pre-subsidy marginal cost	MC_{it}	K_i, k_i
P_{it}^*, p_{it}^*	reset price	P_{it}^*	—
δ_i	Calvo reset probability	δ_i	—
κ_i	sectoral Phillips-curve slope on $mc - p$	— (implicit)	—
$\hat{\delta}_i$, Δ	$\kappa_i/(1+\kappa_i)$, its diagonal matrix	$\hat{\delta}_i(\delta_i, \rho)$, Δ	—

Symbol	Meaning	R23	KMNT
K	$\mathrm{diag}(\kappa_i) = \Delta(I-\Delta)^{-1}$	—	—
$\pi_t, \pi_t^C, \pi_t^\varphi$	sectoral, CPI, index inflation	$\pi_t, \pi_t^C, \pi_t^\varphi$	—
φ	inflation-index weights	φ	—
$y_t^{nat}, l_t^{nat}, \tilde{y}_t$	natural output, natural hours, output gap	$y^{nat}, l^{nat}, \tilde{y}$	—
$Q \equiv (I-\Omega)(I-\mathbf{1}\beta^\top)$	relative-price operator	— (implicit)	—
$\chi_t \equiv p_{t-1} + L\log A_t$	Phillips-curve state	χ_t	—
$M \equiv (I+KQ)^{-1}$	inverse pass-through matrix	—	—
$V \equiv MKQ = I - M$	cost-push matrix	V	—
$B \equiv MK\alpha$	gap-to-inflation vector	B	—
$\delta^\varphi \equiv \varphi^\top(I-\Delta\Omega)^{-1}\Delta$	rigidity-discounted exposure of index φ	δ^φ	—
$\delta^\beta\alpha$	wage pass-through into CPI (scalar)	$\delta^\beta\alpha$	—
κ^φ	slope of index-φ Phillips curve	κ^φ	—
u_t^φ, u_t^C	cost-push residual of index φ, of CPI	u_t^φ, u_t^C	—
v^C	CPI exposure vector (5.20)	— (implicit in eq. 25)	—
r_t^n	natural real rate	r_t^{nat}	—
$S \equiv$ $\Omega_{CL} + \Omega_{CF}\hat{\delta}_F + \Omega_{CD}\hat{\delta}_D$	toy-economy wage pass-through (design doc §7)	$\alpha_F + (1-\alpha_F)\delta_I$ in Example 3	—

Symbols deliberately *not* used, to avoid clashes

Avoided	Reason
ε for the CES elasticity (R23)	reserved for productivity (KMNT); CES is θ
μ for the markup (R23)	reserved for the mean of ε (KMNT); markup is $\mathcal{M}$
α for the IO matrix (KMNT)	reserved for R23's direct labor shares; IO matrix is Ω
ω for Domar weights (KMNT)	R23's λ is used
λ for the log SDF (KMNT)	conflicts with Domar weights; log SDF weight is $\tilde{\lambda}$
θ for production elasticities (R23)	not needed (Cobb–Douglas); reused for CES
γ_i, χ_{ik} for KKT multipliers (KMNT)	conflict with risk aversion and χ_t; multiplier is ν_i
ρ for risk aversion (KMNT)	R23's ρ (discount factor) kept; risk aversion is γ